Carolina University

TITLE OF PROJECT

The Future of Youth Tobacco Use: Predictive Clustering and Time series Analysis

by

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The Future of Youth Tobacco Use: Predictive Clustering and Time series Analysis

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I. ABSTRACT

This capstone study explored and anticipated US youth tobacco consumption. Past data was analyzed using Linear Regression and ARIMA models to predict future trends. Between 2018 and 2020, youth consumed less tobacco, according to Linear Regression. Empirical data showed higher consumption rates, suggesting that external factors including new tobacco products, changing social norms, and the COVID-19 pandemic may have caused discrepancies. The analysis undertook clustering research to better understand regional tobacco use differences. This study grouped states by tobacco usage. clustering analysis revealed separate clusters with varying tobacco use levels, highlighting geographical variation in young people's tobacco use.

Keywords: tobacco use, youth, predictions, clustering, Linear regression, ARIMA, health, smoking, e-cigarettes, data analysis, machine learning, clustering, trends, external factors, statistical modeling

II. BACKGROUND

Most people start smoking before they turn 18. Adding flavors to the goods might make them more appealing to teens and young adults. However, most of the students (80.2% of high school students and 74.6% of middle school students) said they had used flavored tobacco products in the last 30 days. Most high school (90.3%) and middle school (87.1%) students who used e-cigarettes in 2023 said they used flavored e-cigarettes. E-cigarettes, also known as electronic cigarettes, are often just called "e-cigs" by random people. Since 2014, they have slowly become the most popular type of tobacco among young people. It was said that one in 22 middle school students would use electronic cigarettes by 2023. In the last 30 days, 22 middle school students have used them. Some of the tobacco products that were used were cigarettes, cigars, oral tobacco, hookahs, hot tobacco products, and nicotine puffs. Most of those students had used them in the last 30 days.

It was mostly made up of kids and young people who used different kinds of tobacco products. About 2.5% of middle school students and 3.9% of high school students said they had used more than two tobacco products in the 30 days before the poll. Also, teens still use a lot of different kinds of tobacco products, and when they become hooked to nicotine, they put themselves in a lot of danger. Most of these will be able to use the same thing forever as adults. If you smoke cigarettes, cigars, smokeless tobacco, or pipe tobacco, those are tobacco products. If you take them by mouth, they are bidis, hookahs, hot tobacco products, and nicotine packs. There should be steps taken to stop young people from starting to use tobacco products in the first place if we want to lower the number of young people who smoke in the country. How much tobacco teens use relies on a lot of things, such as their social and physical surroundings, their biology and genetics, their mental health, and how they feel about using or not using tobacco products. A young person may start smoking if they have a low income, parents who can't support them, the product is easy to get, it's expensive, they do badly in school, they don't like themselves, or they see ads for tobacco products. National, state, and neighborhood programs can help cut down on the number of young people who smoke. Some of the things that might be part of the programs are moves to make tobacco products more expensive, more places to smoke being banned, raising the minimum age to buy tobacco, and encouraging people to live their lives without smoking. Religion with greater involvement, race pride, and very high academic success are some of the other things that should be considered as the number of young people who smoke goes down. (***2023)***

III. INTRODUCTION

The Data Science Lifecycle is an all-encompassing manual comprising methodical guidelines, strategies, and approaches that facilitate the application of machine learning and other analytical techniques to data to extract insights and predictions that are crucial for achieving business objectives. Essentially, this is the process by which model evaluation follows a series of subprocesses that are executed sequentially, beginning with data preparation, purification, and modeling. This is a lengthy procedure that will require several months to complete. (Coursera, *Data Science Life Cycle*)

The objective of this research is to examine the prevalence of tobacco use among young individuals. A cluster and time series analysis are implemented. This study seeks to investigate tobacco use among youth. It makes use of cluster and time series analysis. This data set entails information on a survey conducted to establish the number of young Americans taking up cigarette smoking and the trend of the numbers by grade over the years. The survey was conducted and collected by the Department of Health and Human Services and taken from data.gov. However, the use of tobacco by the youth has over time reduced, and this is what this dataset represents. Several variables that include the year, place, gender, race, education level, and proportion of cigarette smoking are, however, taken into consideration. One of our prime objectives is to classify the states based on the percentage of cigarette consumption. Using this clustering, one will be able to find which states have high rates of youth tobacco usage and which states have low rates of the same. Such information becomes essential in ascertaining the kind of disparities in cigarette consumption and geographical differences. With this data in hand, that points to public health initiatives and policymaking that could target each region more precisely. Attention will now shift to the change in tobacco use by the youth over the next three years, immediately after the completion of the clustering analysis. Major models used towards this end are the ARIMA model and the Linear Regression model, being two majorly distinct and sometimes used time series analytic tools. One would expect that comparing several models is likely to result in the best method of prediction that is in harmony with the patterns observed in the data. It will be important, therefore, that one makes a comparison of the reliability of each of the ways in a way to look at which one of them would be presenting the best results for how many people will smoke in the future. This shall serve as our framework for the whole cycle of data analysis within this project: know the company inside out, cleaning the data, performing exploratory data analysis (EDA), feature engineering, modeling, and finally, visualizing the results. This will, through a systematic way of studying and understanding, also provide accurate predictions and insights on youth tobacco use. Firstly, in the step of preparation, importing the dataset into Jupyter Notebook is prepared for further analysis through manipulation using the Pandas and NumPy libraries. First, the organization and information of the dataset, including the number of rows and columns, data types, were summarized, and finally, this summarized information was used to draw an initial view of the dataset. Identified missing values were later imputed. In this relation, some cleaning of the dataset was done to ascertain that the selected features are representative of the phenomenon under consideration and, of course, reliable. For other features, missing values were replaced with the average value of the column since this is one of the most common practices for numerical data imputation. That was done with respect to consistency and accuracy of our research. We also checked it very carefully against any duplicate entry which may give rise to any potential distortion of results. Further, we remove all unnecessary data that is not relevant to the analysis. Thus, the data type of columns is validated to identify each feature appropriately for analysis. These activities are essentially purifying and enhancing the dataset, which is the ground to do exploratory data analysis (EDA), feature engineering, and modeling. It, therefore, implies that data cleaning should be done meticulously to ensure a proper and accurate determination regarding the in-depth analysis of the tobacco smoking trends and patterns among the youth.

IV. LITERATURE REVIEW

A. *Case study 1*

An analysis was conducted to determine the factors that predict the density of tobacco outlets across the entire country. The authors of the study are Daniel Rodriguez, Heather A Carlos, Anna M Adachi-Mejia, Ethan M Berke, and James D Sargent. The article "Tobacco control 22 (5), 349-355, 2013" was published in the journal Tobacco Control in 2013 This paper investigated the association of demographic characteristics of US census tracts with the density of tobacco outlets (TOD). Approach: The authors conduct a census tract-level cross-sectional study testing the relationship of socio-demographic factors from the US Census to the density of tobacco outlets in all 64,909 census tracts of the continental USA. Using codes from the North American Industry Classification System, the addresses of retail tobacco outlets were established, and thereby, the density of the outlets was calculated by measuring the number of outlets with regard to tobacco retail outlets for every tract per 1000 individuals. The separate independent factors in analysis were urban/rural classification, percent black, percent Hispanic, percent women of low education level, percent poor families, and median number of persons per household. Through a multivariable analysis, differences in total occupancy density (TOD) per 1000 persons of the urban and rural locations indicated a higher value among the former. For additional evidence, high TOD was highly associated with many African American percentages, Hispanic percentages, less-educated women, and small household sizes. This is a critical difference; both effects are related to the demographic effect, in urban and rural areas, for all socio-demographic groups other than poverty, and the effect is ten times larger for urban census tracts for Hispanics. The results are, therefore, indicative of the fact that the density of such tobacco retailers is more clustered in those regions where such people are located who have more chances and are sensitive to being affected by some adverse health-related effects. Future research should examine TOD with the smoking behavior and measures of smoking cessation in prevalence of illness. (***Daniel Rodriguez, Heather A Carlos, Anna M Adachi-Mejia, Ethan M Berke, and James D Sargent)***

B. *Case study 2*

This study was conducted by Lai et al. The articles aimed to develop prediction models using machine learning algorithms to predict the outcome of smoking cessation. Data was collected from patients enrolled in a smoking cessation program at a medical center in Northern Taiwan, with 4875 enrollments fulfilling inclusion criteria. Models with artificial neural network (ANN), support vector machine (SVM), random forest (RF), logistic regression (LoR), k-nearest neighbor (KNN), classification and regression tree (CART), and naïve Bayes (NB) were trained to predict the final smoking status of the patients in a six-month period. The ANN model achieved a slightly better performance, with a sensitivity of 0.704, a specificity of 0.567, an accuracy of 0.640, and an ROC value of 0.660 (95% confidence interval (CI: 0.617–0.702). The primary outcome of the model was a binary variable (1, 0) defined as the final abstinence status available from the patient within 6 months. The ANN model had the best accuracy of 0.640 and an ROC value of 0.660 (95% CI: 0.617–0.702). The application of the machine learning model was made for physicians and patients to utilize the model. Limitations in the study include the outcome obtained through self-report by the patient, repeated enrollments, and the inclusion of other known and unknown predictors of smoking cessation from the input features**. (Lai et al., 2021)**

A. *Features*

YEAR: The data spans various years, with entries from 2004, 2008, 2015, etc.

LocationAbbr and LocationDesc: These columns provide geographical information, such as state abbreviations (OH, AL, WV, IL, CT) and full state names (Ohio, Alabama, West Virginia, Illinois, Connecticut).

TopicType: All entries in the first few rows are categorized under "Tobacco Use – Survey Data."

TopicDesc: Descriptions vary, including "Cigarette Use (Youth)" and "Smokeless Tobacco Use (Youth)."
MeasureDesc: This column describes the measure being reported, such as "Smoking Status" and "User Status."
DataSource: The source of data for all entries is "YTS" (presumably Youth Tobacco Survey).
Response: The responses include categories like "Ever" and "Frequent," indicating the nature of tobacco use.
Data_Value_Unit: The unit for the data values is "%," indicating percentages.
Data_Value_Type: Type of data value is "Percentage."
Data_Value: This column would contain numerical data indicating the percentage values of the measures,
GeoLocation: Geographic coordinates for the locations are provided, allowing for mapping or spatial analysis.
Gender, Race, Age, Education: These columns contain demographic information, though specific values. (2023)

V. PRE-PROCESSING AND CLEANING

The dataset was uploaded to the notebook and then imported libraries for data manipulation and analysis to get the data ready. As a first step in getting the data ready, we counted the rows and columns in the dataset. To keep the data intact, this phase got rid of duplicate entries. After looking for missing values, the dataset is mostly devoid of two characteristics. The lack of sufficient data for analysis led to the elimination of these traits. The values of the 520 'Response' columns, among others, were filled up via imputation since they were missing. For the sake of completeness and continuity in the dataset, we substituted the means of the columns for missing numerical values and 'Unknown' for missing 'Response' values. After we finished cleaning, we looked at the data structure to see how diverse and distributed the values were for our features. The variety and breadth of our data were demonstrated by its unique numerical and category values. Thorough statistical examination of numerical aspects concluded our data preparation. We demonstrated the distribution and central tendency of our data by calculating and analyzing the mode, first, second, and third quartiles, as well as the mean, median, and mode. We were ready to go into research and modeling after reading this comprehensive statistical overview, which put our dataset's numerical context into clearer perspective.

VI. VI EXPLORATORY DATA ANALYSIS

1) Univariate analysis

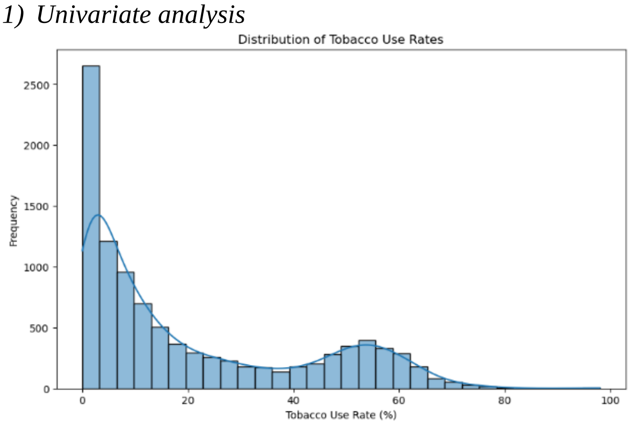


Figure 1: distribution of tobacco rate.

Following data preparation, our project's primary emphasis was univariate analysis, with a focus on the 'Data Value' column—representing rates of tobacco use. We used box plots and histograms to examine the distribution and found that the data was right skewed with a few outliers; this meant that although many people used tobacco at modest rates, there were a few who used it at much greater rates. For a comprehensive picture of the teen tobacco use patterns in our dataset, this study was vital.

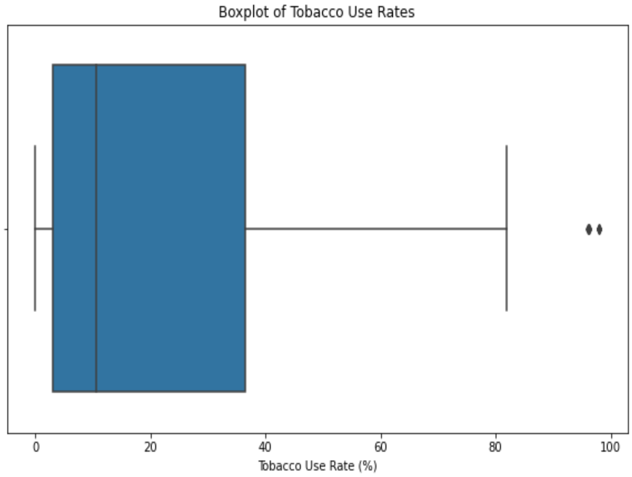


Figure2: boxplot.

The figure represents the box plot of tobacco use rate % from the survey dataset. The median tobacco use from the plot shows around 20%. The overall range starts from 0-80 percent. The box is slightly skewed to the right which also tells from the dataset the survey found that most of the data have low tobacco rates.

2) Bivariate analysis

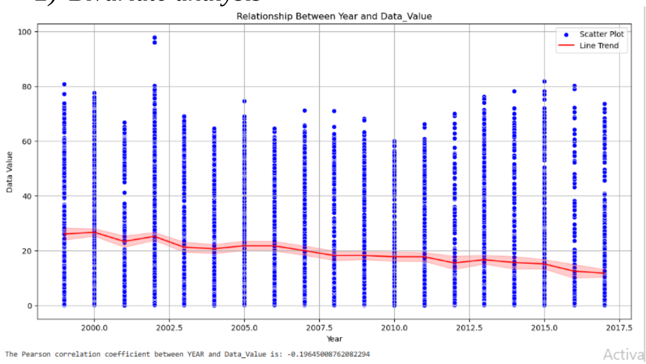


Figure 2:relationship between year and tobacco use rate.

We thoroughly investigated tobacco usage patterns and trends among youth during Exploratory Data Analysis (EDA). We initially calculated the annual tobacco use average by merging the data to find the mean. We added geographic areas to this method to see tobacco use trends and differences. Histograms and other visualizations showed patterns and distributions accurately. We divided the data by gender and education to analyze it more thoroughly. This allowed us to study these demographic groups' modest tobacco consumption disparities. One of the key visuals showed which gender used tobacco more and which type they used. We also examined the most popular tobacco products among youth. The below mentioned are average of tobacco use by year.

	Year	Average tobacco use (%)
	1999	26.07
	2000	26.6
	2001	23.32
	2002	25.1
	2003	21.24
	2004	20.68
	2005	21.74
	2006	21.71
	2007	20.02
	2008	18.32
	2009	18.35
	2010	17.9
	2011	17.88
	2012	15.94
	2013	16.97
	2014	16.13
	2015	15.41
	2016	13.7
	2017	12.66

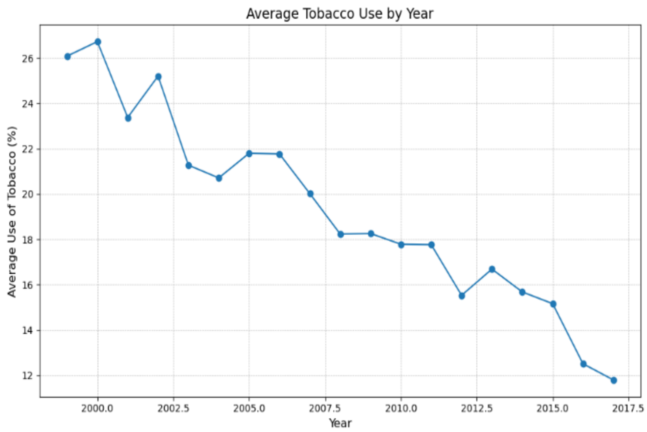


Figure 3:average use of tobacco by year.

The line chart shows that there is a trend of change over time shown by the line chart, which shows the average amount of tobacco used by year. Between 1999 to 2017, the average number of youths who used tobacco dropped from almost 26.60% in 1999 to about 12.66% in 2017 when it was at its lowest point. Based on this trend, it looks like youth have been smoking less in general over the years.

Average tobacco uses by location/state.

Locations	Average tobacco use (%)
Alabama	24.11456
Arizona	16.41107
Arkansas	26.04402
California	21.45208
Colorado	24.70217
Connecticut	15.93342
Delaware	17.90609
District of Columbia	19.82791
Florida	29.02604
Georgia	20.47099
Guam	31.23125
Hawaii	14.12151
Idaho	20.10217
Illinois	17.81801
Indiana	18.98015
Iowa	20.02923
Kansas	18.03429
Kentucky	28.22039
Louisiana	21.10704
Maine	19.29787
Maryland	22.95729
Massachusetts	21.40227
Michigan	23.09783
Minnesota	16.55799
Mississippi	20.53528
Missouri	17.98792
National (States and DC)	60.53667
Nebraska	23.02879
New Hampshire	18.33353
New Jersey	16.80561
New Mexico	20.9125
New York	23.49111
North Carolina	19.41114
North Dakota	20.3902
Ohio	16.82667
Oklahoma	22.74375
Pennsylvania	18.64156
Puerto Rico	23.16667
Rhode Island	21.02014
South Carolina	20.21376
South Dakota	21.37807
Tennessee	28.1756
Texas	27.88542
Utah	13.01797
Vermont	19.31209
Virgin Islands	6.166667
Virginia	7.507895
West Virginia	25.17419
Wisconsin	19.16586
Wyoming	27.38333

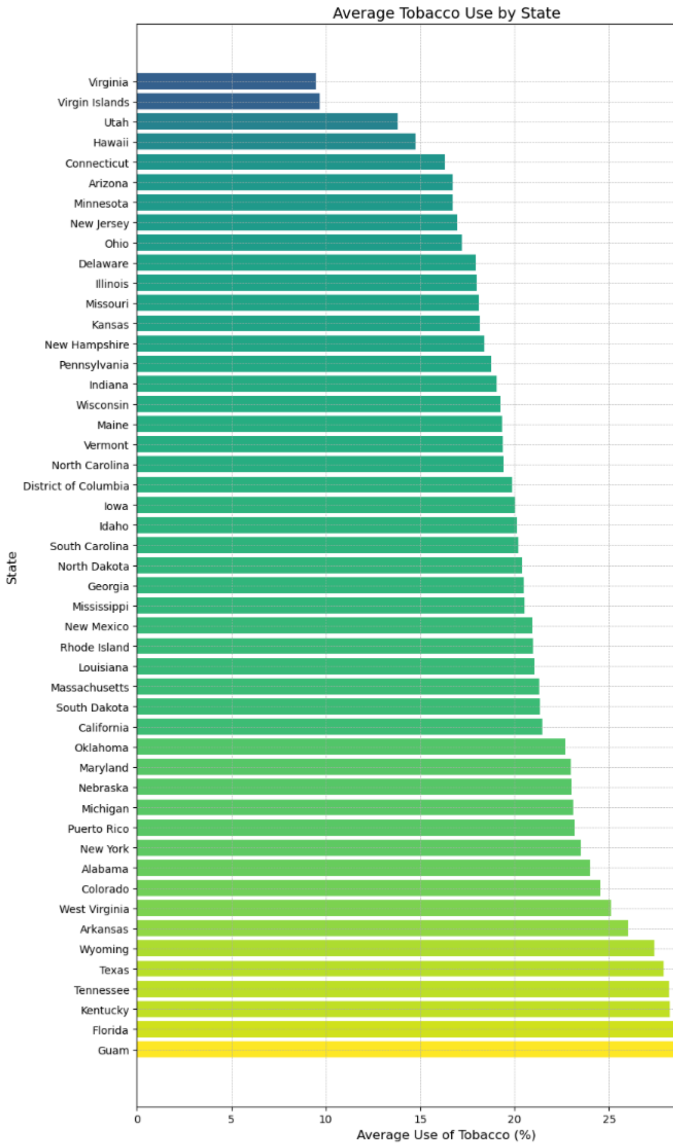


Figure 4:tobacco use by location.

The bar chart depicts the mean tobacco consumption across different regions, which are the states of the United States of America. States are listed in ascending order on the chart. Based on the graph, it is evident that Florida has the highest average tobacco use rate at 29.02%. In contrast, Virginia has the lowest average tobacco use rate at 9.50%. In the unlikely situation, the chart also presents the mean tobacco percentage for the complete United States, including Guam, a territory of the USA that is not a state but is nonetheless referenced. The chart shows a wide range of average tobacco use percentages among the states, indicating variability in how common tobacco use is among youth across the United States.

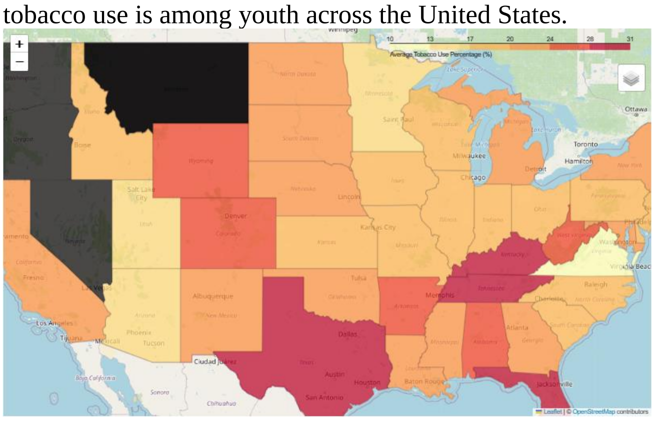
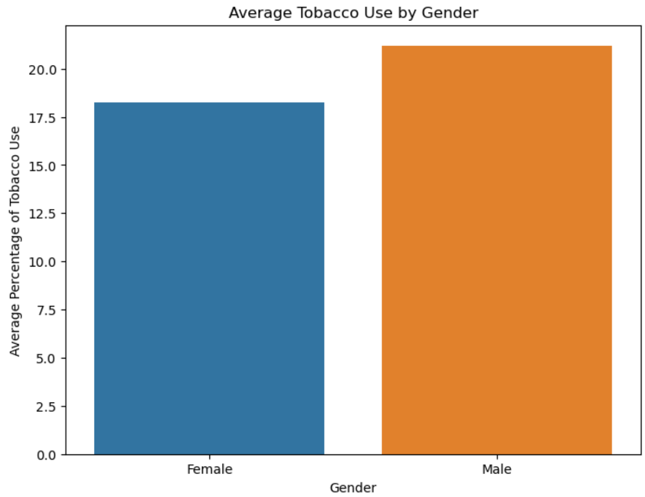


Figure 5:state with color gradient in terms of tobacco use.

Figure 6:tobacco use by gender.



The visualization above represents the average percentage. The mean percentages of tobacco consumption by gender are as follows: The average tobacco consumption for males is roughly 21.15%, while females use tobacco at a rate of about 18.36%. This data indicates a small difference in tobacco usage between males and females.

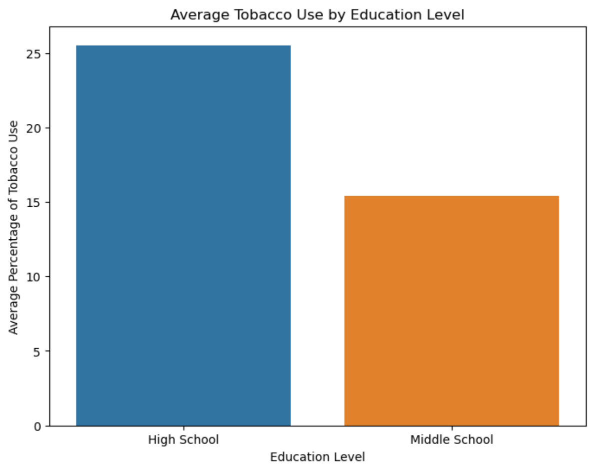


Figure 7: tobacco use by education.

The visual representation shows that the maximum number of students who are in high school are involved in tobacco use compared to middle school. High school students have an average tobacco use rate of approximately 25.47%, whereas middle school students have a rate of approximately 15.81%, according to the data.

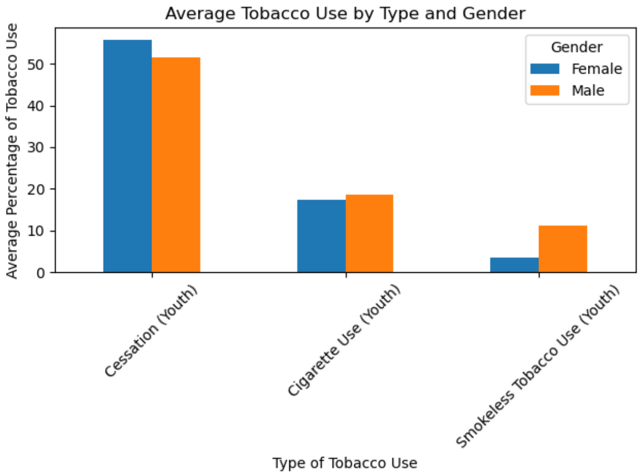


Figure 8: tobacco use by type and gender.

This visualization provides a clear comparison of how tobacco use preferences and behaviors vary between genders for various types of tobacco products. The study showed that about half of the young people who took part in programs to quit smoking were girls. This is more than boys. A little more boy than girls smoke cigarettes. Overall, about 18% of teens and young adult's smoke. But when it comes to tobacco products like snuff or chewing tobacco, boys use them a lot more than girls. Only 3% of girls use these products, while 11% of boys do.

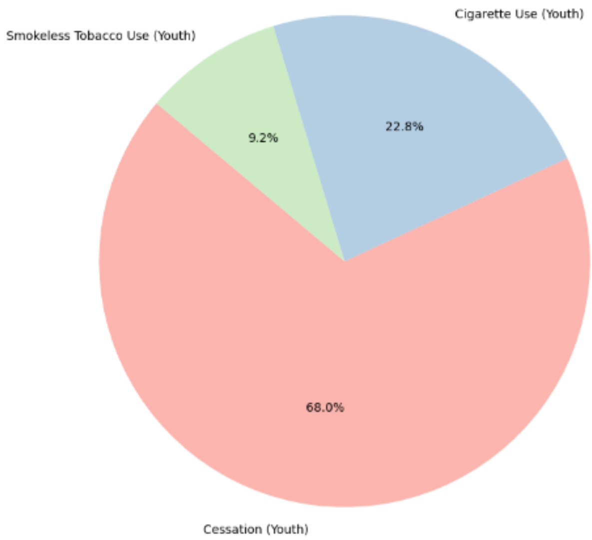


Figure 9: tobacco use rate by topic type

Cessation among youth contributes 68% to the pie chart titled "Average Use of Tobacco Products by Youth." Use of cigarettes (Youth), 22.8%; Use of smokeless tobacco (Youth), 9.2%. Indicates the largest part shows most of the youth do not use tobacco or are trying to quit.

VII. FEATURE ENGINEERING

Feature engineering is an essential step in data analysis that involves extracting valuable insights and patterns. By calculating the mean tobacco usage for each state, we can categorize states with comparable rates and distinguish unique tobacco consumption patterns. Linear Regression and ARIMA models are employed for time series forecasting, specifically targeting the mean tobacco consumption each year. This facilitates the comprehension of patterns in tobacco consumption over a period, allowing for future projections and evaluation of their consequences.

VIII. METHODOLOGY

A. K means algorithm

The K-means algorithm is a centroid-based clustering method in which the distance between each data point and a centroid is utilized to group the data into clusters. Determining the Kth number of categories within the dataset is the objective. K-means clustering, which originated in signal processing, is a vector quantization method that divides n observations into k clusters. Each observation is assigned to the cluster whose average is closest to it, which functions as the prototype for the cluster. The process of designating groups to individual data points is iterative, and data points are progressively clustered according to shared characteristics. Minimizing the aggregate of the distances between each data point and the centroid of the cluster is the goal to determine which group each data point should be assigned to. *(Sharma, 2023)*

In our capstone assignment, we implemented K-Means clustering of the States in the United States by their mean tobacco consumption. This method has thus provided a way of grouping patterns and similarities concerning the usage of tobacco in different geographical locations. Initialization stage of K-Means: The first step in the clustering process was undertaken in the determination of the centroids; with the help of the elbow method, it was three, denoted K. After these centroids were arbitrarily taken from the dataset, the next step emerged in the process of clustering. The k-means algorithm has the equation down there.

$$\sum_{j=1}^{k}\sum_{i=1}^{n}\left\|x_i^{(j)}-C_j\right\|^2$$

In this phase, each state from the data has been assigned to the nearest centroid; hence, discrete clusters form three. The following was computing the Euclidean distance between the centroids and mean tobacco use value of every state in a cluster. Updating of the centroids of each cluster after the reassignment is made. Thus, the position of the centroid has been recalculated in a way that could represent more accurately the clusters of aggregation of the state's tobacco use values designated to it. Each assignment and update iteration were refining the centroid locations and cluster assignments. The algorithm converged when centroids became stabilized, and the final clusters were formed with centroids whose position changes did not have a great difference; it pursued its operation until the centroids reached stability.

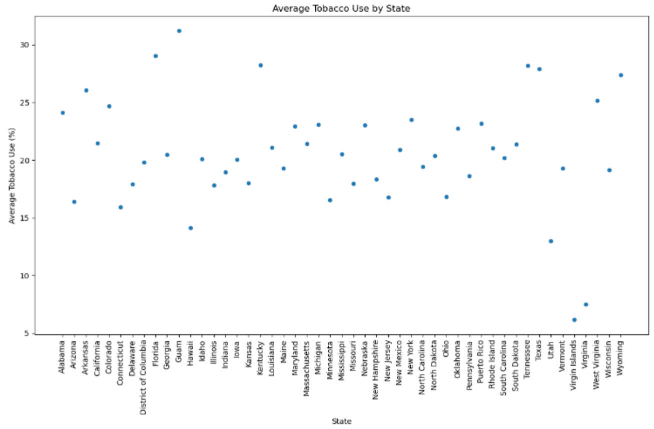


Figure 10:actual datapoints of tobacco use by state.

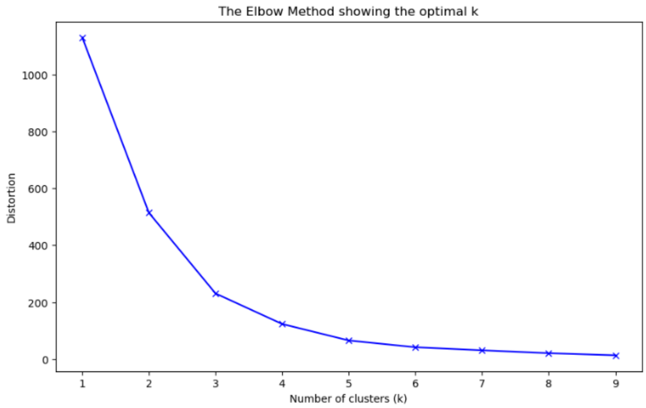


Figure 11:elbow method to check optimal number of clusters.

The K-Means algorithm would investigate minimizing the sum of squared distances between the data points (state) and the respective cluster centers. This is the definition of the objective function J used by K-Means. The main variable that comes into play in the state_tobacco_use features, at the stage of preparing the data, is the average level of tobacco use. The elbow method plot would therefore suggest that 3 was the appropriate number of clusters, given the variation from the rate of decrease within the sum of squares of the clusters. We applied a K-Means clustering with the parameter K = 3 in dividing the states into three discrete subgroups according to the levels of tobacco consumption. Post-clustering, analysis is done on these clusters to find insight into the distribution of tobacco use. A cluster is a group of states that influence similar levels of tobacco use, represented first and then interpreted to find patterns and similarities regionally. An elegant framework is provided in this article that explains the entire US tobacco use landscape based on the K-means algorithm

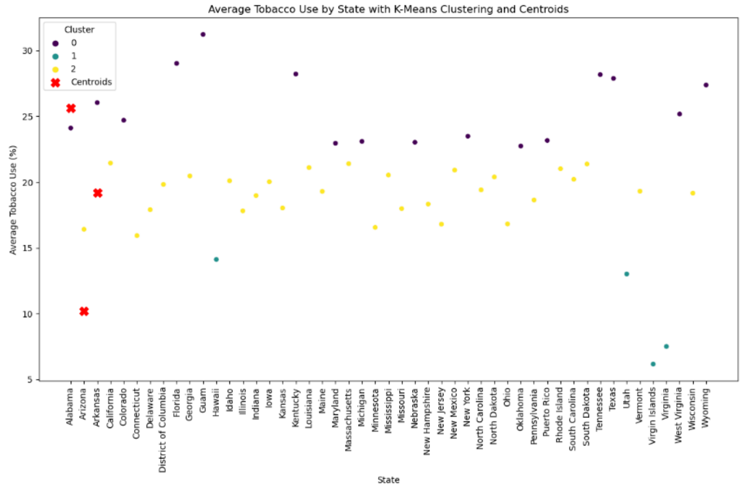


Figure 12: clusters with its centroid.

B. *Linear regression*

Linear regression predicts the relationship between two variables on the assumption that the independent and dependent variables are connected linearly." It attempts to identify the line where the sum of squared differences between predicted and actual values is minimized. This method is utilized in numerous fields, including finance and economics, to forecast and analyze data trends. Simple linear regression involves just a single dependent variable and an independent variable. The model calculates the intercept and slope of the line of best fit, which symbolizes the correlation among the variables. When the independent variable is set to zero, the intercept represents the predicted value of the dependent variable. The slope, on the other hand, indicates the change in the dependent variable for each unit change in the independent variable. In machine learning, linear regression is the most straightforward and silent statistical regression technique utilized for predictive analysis. It is known as linear regression when the X-axis represents the independent (predictor) variable, and the Y-axis represents the dependent (output) variable. Linear regression is considered simple linear regression when a single input variable, denoted as X (the independent variable), ***(Mali, 2024)***

We followed a methodical procedure to forecast tobacco use for the years 2018, 2019, and 2020 utilizing linear regression. The data in our dataset comprised annual observations of tobacco use among middle and high school students. Every individual entry in the dataset denoted the mean percentage of tobacco consumption for a particular year. Linear regression is a statistical technique used to analyze the connection between a dependent variable (y) and one or more independent variables (x).

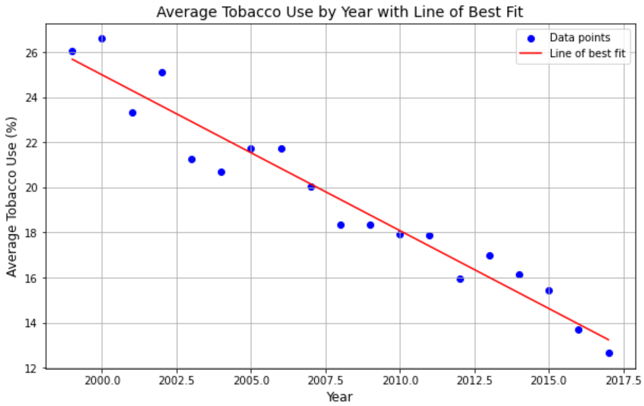


Figure 13: best fit line between year and tobacco use.

In our study, we considered the proportion of tobacco usage as the dependent variable, whereas the year was treated as the independent variable. The connection is defined by the linear equation:

$$y=mx+c$$

In this context, the variable y represents the estimated percentage of tobacco use. The variable m represents the slope of the line, indicating the rate of change in tobacco use each year. The variable x represents the year being considered. Lastly, the variable c represents the y-intercept, which represents the initial level of tobacco use at the start of the time series when x is equal to 0.

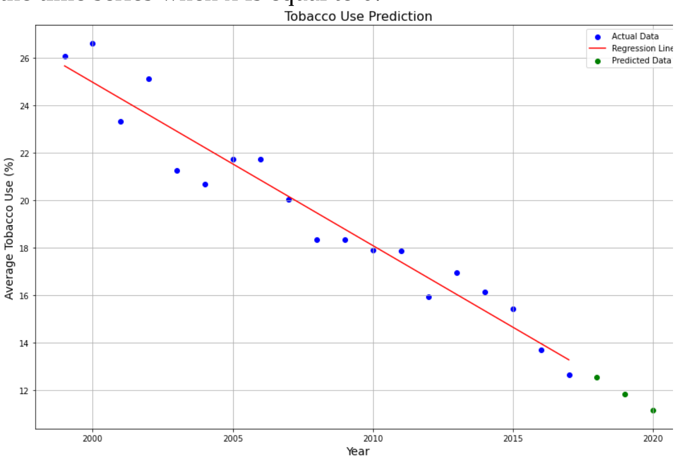


Figure 14: predicted points.

We utilized historical data up till 2017 to train the Linear Regression model. During this phase, the model algorithm determined the optimal line that best fits the data points by changing the variables m and c to reduce the discrepancy between the observed and estimated levels of tobacco consumption. This procedure is commonly performed using a technique known as "least squares," which seeks to minimize the total of the squares of the residuals (discrepancies between observed and predicted values).

After completing the training of the model and obtaining the line equation, we utilized it to make predictions for future values. We entered the years 2018, 2019, and 2020 into the algorithm to obtain the projected percentages of tobacco usage for those years. To compute the expected tobacco usage, we replaced the variable x with the value 2018 in our line equation for the year 2018. The analysis generated forecasted estimates of tobacco consumption for the upcoming three years, utilizing the past pattern as a basis. These forecasts were based on the premise that the rate of change in tobacco usage would stay stable over time, reflecting the ongoing trend found in the historical data.

Using Linear Regression, we employed a simple yet potent technique to predict future patterns by analyzing historical data. This methodology enabled us to measure the anticipated shifts in tobacco consumption among high school students and middle school students enabled us to establish a starting point for assessing the real-time patterns as they developed.

C. ARIMA model

ARIMA stands for "Autoregressive Integrated Moving Average," and this is an abbreviation which by far can be considered one of the most widely used statistical techniques applied in time series data forecasting. It studies the different patterns with respect to past data and relates to the patterns in future prediction. (***Hayes adam)***

The ARIMA model is a forecasting method that we utilized in our capstone project to predict youth tobacco usage. All values of the ARIMA model are determined by three parameters (p, d, q). The values the ARIMA parameters take are as follows: p (autoregressive order)—it indicates the number of lags to be used in the prediction of the differenced variable; (integrated order)—it shows the number of differences to obtain stationarity of the time series. It has a constant mean. And variance throughout time. The "q" shows the order of the Moving Average model. It shows what lags the forecast error variables are.

$$\left(1-\sum\nolimits_{i=1}^{P} \phi_i L^i\right)(1-L)^d \;\; Y_t = \left(1+\sum\nolimits_{i=1}^{q} \phi_i L^i\right)\varepsilon_t$$

The time series is denoted by Y. The autoregressive component consists of the parameters. Φi. The parameters of the moving average component are being referred to. θi. noise errors are $\mathcal{E}t$. The lag operator is denoted by L

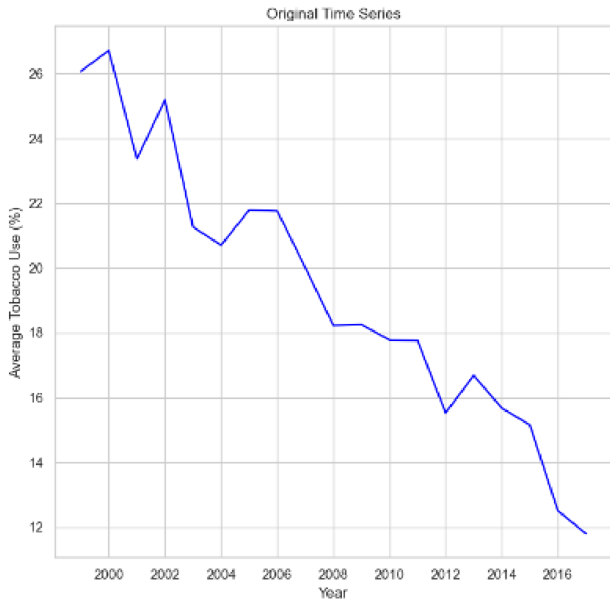


Figure 15:actual time series data.

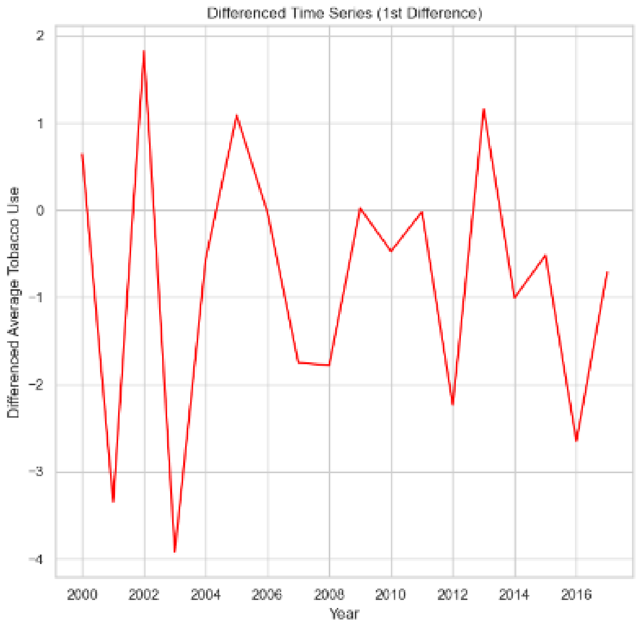


Figure 16:differenced time series.

For this capstone project, the ARIMA (Autoregressive Integrated Moving Average) model was used in predicting the prevalence of tobacco use among the young. So, we first need to test the stationarity of our time-series data representing yearly percentages of tobacco usage. Where data in parameter d appeared non-stationary, it was differenced. to have that condition satisfied. Stationary data should be utilized in ARIMA modeling.

The model involves the autoregressive (p) and the moving average (q) orders of the respective model, with the help of autocorrelation function (ACF) and partial autocorrelation function (PACF) plots. These plots were helpful in finding the lag points in such a way that partial autocorrelations effectively die; hence, they helped in finding the appropriate values for p and q.

ARIMA parameters

P	*1*
d	*1*
q	*0*

This way, setting the values for p, d, and q, the time series data was applied in the ARIMA model. A model fit was made by estimating the best autoregressive (ϕ) and moving average (θ) coefficients that will reflect the patterns and dynamics in the historical data. Once the model was fitted, it was possible to project the usage of tobacco percentages for these years: 2018, 2019, and 2020. All this was done with the patterns found in historical data and, of course, with the structure of the ARIMA model.

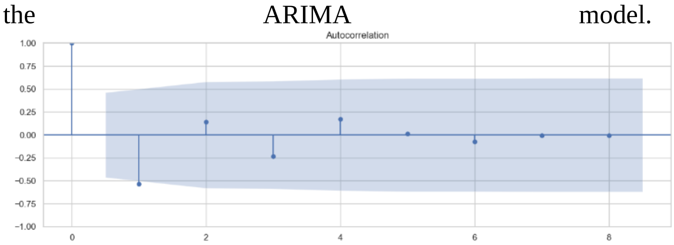


Figure 17: autocorrelation.

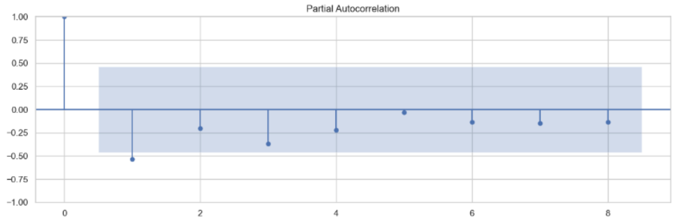


Figure 18: partial autocorrelation.

This determines the performance evaluation of the model based on statistical measures such as Mean Absolute Error (MAE) and Mean Squared Error (MSE) to determine R-squared (R^2) values that tell whether the model makes good predictions. These three measures provide an arithmetic foundation for assessing the relevance and reliability of the trends forecasted by the model for future behavior in the use of tobacco by juveniles. All these were done under a situation in which projections are available for comparison with the forecast and actual values.

IX. Result

a) Clustering Outcomes

Cluster	**Mean Tobacco Use (%)**	**Locations**
0	25.65%	Alabama, Arkansas, Colorado, Florida, Guam, Kentucky, Maryland, Michigan, Nebraska, New York, Oklahoma, Puerto Rico, Tennessee, Texas, West Virginia, Wyoming,
1	10.20%	Hawaii, Utah, Virgin Islands, Virginia
2	19.18%	Arizona, California, Connecticut, Delaware, District of Columbia, Georgia, Idaho, Illinois, Indiana, Iowa, Kansas, Louisiana, Maine, Massachusetts, Minnesota, Mississippi, Missouri, New Hampshire, New Jersey, New Mexico, North Carolina, North Dakota, Ohio, Pennsylvania, Rhode Island, South Carolina, South Dakota, Vermont, Wisconsin

Cluster 1 comprises states such as Hawaii, Utah, Virgin Islands, Virginia the states in this cluster have significantly lower rates of tobacco usage. With the centroid of 10.20%
Cluster 0 includes states with an average tobacco usage rate of 25.65%, such as Alabama, Arkansas, Colorado, etc. This shows that the tobacco usage rate is highest among all the states.
Cluster 2, which represents the state with tobacco use of moderate displays a significantly higher rate than cluster 1 but lower than cluster 2. Some of the states on cluster 2 are: Arizona, California, Connecticut, Delaware. This could be due to the high tobacco usage in specific states or regions not included in the clusters. Evaluation of Current Trends: There

is a lot of regional variation in tobacco use, as seen by the clustering. The lowest tobacco use rates are found in Cluster 1 states, which may be due to efficient public health initiatives, cultural influences, or socioeconomic circumstances. Consequences for Public Health: Cluster 2 has a higher average, suggesting that public health initiatives may need to step up their game to curb tobacco consumption there.

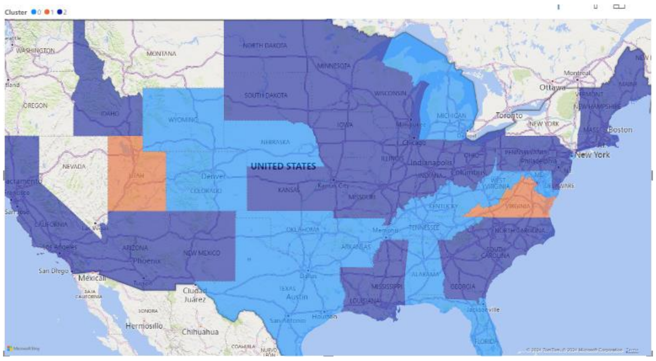


Figure 19:clustered location with color.

The diagram illustrates a United States map wherein the states are categorized into clusters. The pale blue color in the cluster represents states where teenage tobacco use has been elevated by 25.65% over the years. An analogous manner. Cluster 2, represented by the dark blue color, exhibits a moderate prevalence of tobacco use at 19.18% after clustering. The cream state designation denotes the states where adolescent tobacco use is least prevalent, at 10.20%. However, not every state was included in the dataset, so Hawaii and Puerto Rico are also obscured in the figure.

b) Linear Regressions results

Metric	**Value**
Mean Absolute Error (MAE)	0.55%
Mean Squared Error (MSE)	0.344
R-squared (R^2)	0.73%
Forecasted Avg Tobacco Use 2018	12.55%
Forecasted Avg Tobacco Use 2019	11.86%
Forecasted Avg Tobacco Use 2020	11.16%

Model Effectiveness: The Linear Regression model, therefore, performed quite well in general prediction performance of the dataset, with a Mean Absolute Error (MAE) of 0.55% and Mean Squared Error (MSE) of 0.34 The model's predictions are comparable to the actual data points, as evidenced by the low error values. That is, relatively high R-squared value ($R^2 = 0.73$) effectively accounted for 73% of the variance in the historical tobacco use data, further validating the effectiveness of the model. The model, in relation to the prospective trajectory, predicts the constant decline of the mean of tobacco consumption at 12.55% in 2018, 11.86% in 2019, and 11.16% in 2020. From the linear regression analysis, this forecast would mean that, in fact, there was a trend of continuous decline of tobacco consumption.

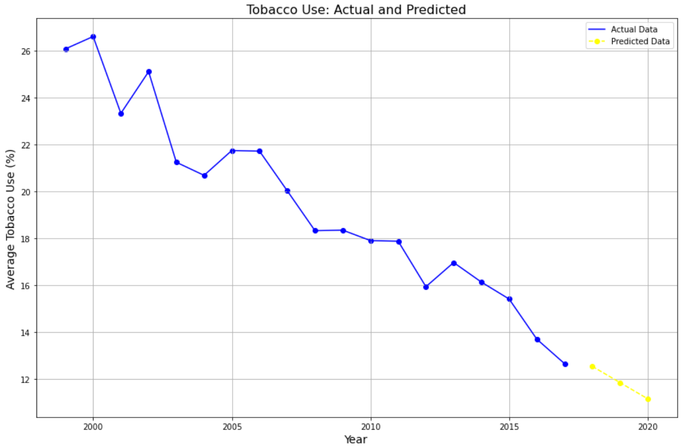


Figure 20:linear regression prediction.

c) ARIMA model results

Metric	**Value**
Mean Absolute Error (MAE)	1.37%
Mean Squared Error (MSE)	2.0270
R-squared (R^2)	-57%
Forecasted Avg Tobacco Use 2018	12.86%
Forecasted Avg Tobacco Use 2019	12.82%
Forecasted Avg Tobacco Use 2020	12.83%

Model Effectiveness

When applied in this dataset, the ARIMA model obtained a Mean Absolute Error (MAE) of 2.0270 and a Mean Squared Error (MSE) of 2.03%. The model exhibited a greater deviation from the actual data points on the higher value side. This indicates there is less precision in the model. The discrepancy is further reflected with a value of -0.57% for the R-squared (R^2), which shows that the ARIMA model was not able to really capture the trend and the variability within the data of tobacco use very well. Consequently, its performance was inferior to that of a basic average-based prediction.

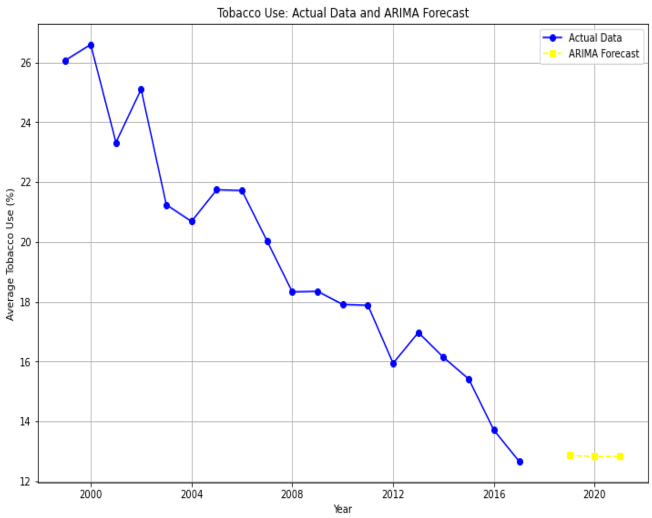


Figure 21: ARIMA model prediction.

X. Comparison of Models and Implications

Based on the above average tobacco uses forecasting results and the forecasts based on ARIMA and linear regression models, one may refer to the following points of differences as a reason for consideration: From the results obtained, Linear Regression gave a better prediction because of its accuracy, with an R-squared value of 0.73 and corresponding MAE and MSE values at 0.55% and 0.34%. It shows how well the model fits the historical data of tobacco use, while at the same time its ability to explain most of the variation in tobacco use over time. The highest rates were recorded in 2018 with an average of 12.55%, while the predicted values for the same in the year 2020 were averaged at 11.16%. These proved to have a linear reduction from past findings, which matched with the fact that tobacco use is still declining.

On the other hand, the error metrics for the ARIMA model did not satisfy their value. For example, in the case of R-squared, it was reviewed at -0.57, mean absolute error (MAE) at 1.37%, and mean squared error (MSE) at 2.03%. The ARIMA model has forecast average consumption rates of cigarettes from 2018 to 2020 at 12.8%, showing that the decline of tobacco use will soon be captured. The comparison of these two shows that the Linear Regression model gave more consistent and accurate predictions of tobacco usage rates in the future years and, at the same time, followed more closely the historical declining tendency than did the Logistic model. On the other hand, falling trends in tobacco usage rates are well captured by the Linear Regression model, while for the ARIMA model, predictions were to remain stabilized. Among the models considered by us for this analysis, that is, ARIMA and Linear Regression, I would henceforth like to refer to and discuss the Linear Regression model in our discussion and the subsequent study. Decisions were made with respect to this, as the model fits well with the trend from the historical patterns pointing towards a decline in tobacco use and it had better performance measures.

The ARIMA model generally fits quite well for most time series analyses; however, in this case, it did not fit as well. It fit less closely to the historically declining trend, though, and predicted that tobacco use rates would level off. Considering the capstone assignment, the derived clearness prevailed that the Linear Regression model was very appropriate for deriving the analysis of the upcoming pattern of tobacco use. It is in line with history, interpretability, and predictive accuracy, which includes this tool as quite reliable for informing strategies and policy.

XI. Discussion

In our capstone project, we used a Linear Regression model to predict rates of tobacco use in youth. And based on the projections, these will have reduced to 12.55% in the year 2018, 11.86% in 2019, and 2020, respectively. Though by all accounts, recorded data will show a considerably higher percentage with a calculated 17.15% in 2018, 21.85% in 2019, and 12.19% in 2020. This is quite a difference against the projected data large enough that it will now have to investigate the possible variables that may have caused such differences

a) Analysis of disparities

The Linear Regression model implies a direct and proportional relationship between time and tobacco usage rates. Now, if this relationship in the real world, under actual circumstances, followed such a linear pattern strictly—or, in fact, it might have followed non-linear patterns at times, alas—then the predictions that this model would make could only be at variance with reality. External Influences: A wide range of external influences, from legislative change through changes to public health advertising, to economic conditions and new promotion of tobacco products (such as e-cigarettes), have major impacts on the likely success for the decline or reduction in smoking rates. Social and cultural norms may lead to changes in the trend: For example, the fast-emerging prevalence of vaping, especially among youths, might have indicated an upward shift in prevalence rates that were observed in the year 2019. Social and cultural trends, therefore, affect the changing use of tobacco through changing social views. Peer influence, social media, and marketing shape behaviors in huge magnitudes that can possibly inspire young adults to use more tobacco, and this was not factored into the linear model. Data Collection and Reporting: The methods used in due course of time to be subjected for data collection and reporting regarding tobacco use can be subjected either to change the quality of data accuracy or its consistency. Differences between the projected and realized rates could occur due to changes in surveys, their frequency, or the sample demographics. Other factors, maybe something like the COVID-19 pandemic, probably contributed to the trends of tobacco use. The social and economic toll of the pandemic, for example, may have caused increased stress levels and patterns of drug abuse change, including that of tobacco.

As a matter of fact, the difference between the Linear Regression model's forecast and actual data clearly points to further confirmation that forecasting any behavior—either tobacco-using or non-tobacco using—is highly complex. On the other hand, the linear models of change assist in providing bases for grasping long-term patterns, though they may not account fully for abrupt changes or complex impacts from some external influences. The conclusion that can be drawn speaks much for the importance of including a wider set of variables and perhaps using more sophisticated models that could adjust to non-linear trends and outside influences to better predict future youth tobacco use trends.

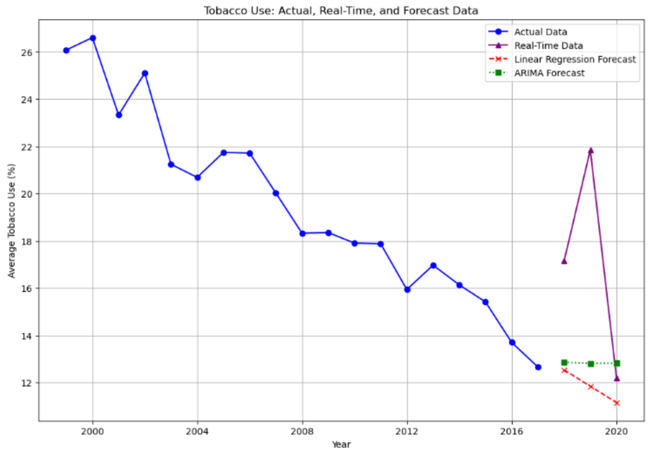


Figure 22: figure comparing models with real time data.

XII. Conclusion

This capstone study involved the use of Delphi for forecasting and cluster analysis into patterns of tobacco use among the youth in the United States. Using the Linear Regression and ARIMA models, it was meant to establish the disparities, geographical location to consume tobacco among states. This depicts declining trends in youth tobacco use through the Linear Regression model, which predicted falling rates of tobacco use for every following year than the previous. The ARIMA model, in this case, could not give the right prediction of such trends. Nonetheless, the clustering drew out some regional variations, and from each of the clusters, there were some states that clearly had much higher rates of tobacco use than other states. Trying to forecast the tobacco use trend among young people is complex, given that plenty of influences, including social norms, laws, and economic issues, may direct the trend. It is pointed out that the huge difference in tobacco uses between the predicted and the actuality, on the other hand, requires pointing out the need for much more sophisticated models that could incorporate nonlinear patterns and external influences to improve prediction accuracy. The study, therefore, focuses not only on some of the most pressing contemporary aspects of youth tobacco use but also on the importance of required and focused public health effort and policymaking aiming to cater to regional disparities to further be able to help fuel the decline prevalent in overall youth tobacco consumption.

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